

Computational Validation of Adaptive Interface Control in TMI Quantum Systems

Расширенный отчёт v4 по вычислительным тестам 1-27

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Аннотация

Этот отчёт обновляет вычислительную линию ТМИ до Tests 1-27. Версия v4 добавляет внешнюю симуляторную ветку Tests 23-27: raw Qiskit-Aer failure, Qiskit noise calibration, calibrated Qiskit-Aer validation и circuit-inverse correctness audit. Главная проверяемая цепочка остаётся узкой:

$$I_Q^{coh-auto} \Rightarrow T_2^{eff} \uparrow \Rightarrow P_{success} \uparrow.$$

Новый внешний результат:

$$Calibrate(I_{sim}) \wedge Correct(I_{circuit}) \Rightarrow P_{TMI} > P_{standard} > P_{base}.$$

Это не лабораторное доказательство физического поля; это воспроизводимая вычислительная и Qiskit-Aer симуляторная валидация прикладного механизма автоинтерфейсного управления.

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1. Статус отчёта

TMI Quantum Computational Validation v0.4

Главная научная особенность v4: внешний Qiskit-Aer перенос сначала провалился, затем был откалиброван, а затем был очищен от ошибки обратной схемы.

2. Summary of Tests 1-27

Test	Name	Criterion / meaning	Result
1	Ramsey toy model	$T2_TMI > T2_standard$	Delta T2 = 58.1 us
2	Noisy Ramsey learning	$T2_TMI > T2_standard$	Delta T2 = 39.3 us
3	Small quantum circuit	$P_TMI > P_standard$	Delta P = 0.115
4	Stabilized interface	Stable adaptive control	Delta P = 0.026
5	Surrogate Monte Carlo	$E[P_TMI] > E[P_standard]$	Delta = 0.0510, Pr = 0.864
6	Ablation study	Structured adaptive control	full TMI Delta = 0.0680
7	Density-matrix Monte Carlo	Full matrix validation	Delta = 0.0930, Pr = 0.933
8	Cross-model validation	Multiple circuit families	Delta = 0.0660, Pr = 0.846
9	Failure-boundary map	Weak/failure regions	failure = 0, weak = 6
10	Adversarial failure	Forced failure modes	failure cells = 58
11	Safety recovery	Mitigation after failure	recovered = 31/58
12	Reproducibility harness	One-command validation	passed = 11/11
13	Independent simulator	Independent simulator validation	Delta = 0.1121, Pr = 1.000
14	Noise-model stress	Noise class robustness	passed = 3/4, Delta = 0.0068
15	Amplitude damping recovery	Relaxation-aware layer	Delta_relax = 0.1532
16	Noise classifier	Classify then control	accuracy = 0.952, Delta = 0.0554
17	Online noise drift	Dynamic reclassification	Delta = 0.0255, acc = 0.816
18	Delayed feedback	Latency hurts online control	online = 0.883, delayed d2 = 0.861

Test	Name	Criterion / meaning	Result
19	Latency policy optimization	Predictive latency layer	opt = 0.875 > d2 = 0.861
20	Cost-overhead tradeoff	Utility and efficiency	U_highperf = 0.0705
21	Cost-interface boundary	B_cost	lambda* = 0.0636
22	Multi-objective control	Boundary manifold	standard share = 0.608, cost-aware-low = 0.373
23	Standard simulator protocol	Backend-independent Qiskit-style protocol	Delta = 0.0414, Pr = 0.750
24	Raw Qiskit-Aer execution	External simulator without calibration	failed, Delta = -0.00097, cells = 0/16
25	Qiskit noise calibration	Simulator-interface calibration	54 candidate calibrations; Loss_sim = 0.00106
26	Calibrated Qiskit-Aer validation	Held-out calibrated validation	passed, Delta = 0.0725, cells = 12/16
27	Circuit-inverse audit	Correct inverse and rerun Qiskit-Aer	passed, Delta = 0.1144, cells = 16/16

3. External simulator branch: Tests 23-27

Test 23 created a backend-independent standard-simulator protocol. The key update in v4 is the real Qiskit-Aer branch:

Test 24 : failed → *Test 25 : calibration* → *Test 26 : calibrated passed* → *Test 27 : correct*

3.1. Test 24: raw Qiskit-Aer failure

$$P_{base} = 0.7535, \quad P_{standard} = 0.7532, \quad P_{TMI} = 0.7523.$$

$$\Delta(P_{TMI} - P_{standard}) = -0.00097, \quad Pr(P_{TMI} > P_{standard}) = 0.0729.$$

This established the need for simulator-interface calibration:

$$I_{sim} : TMI \text{ model} \rightarrow Qiskit \text{ NoiseModel}, \quad Loss_{sim}(I_{sim}) > 0.$$

3.2. Test 25: Qiskit noise calibration

Test 25 found 54 candidate calibrations. In the default-like calibrated regime:

$$\Delta(P_{standard} - P_{base}) = 0.0191, \quad \Delta(P_{TMI} - P_{standard}) = 0.0299.$$

It also found incomplete idle-gate preservation:

$$idle_transpile_id_preservation_rate = 0.4375.$$

3.3. Test 26: calibrated Qiskit-Aer validation

$$P_{base} = 0.4793, \quad P_{standard} = 0.5203, \quad P_{TMI} = 0.5928.$$

$$\Delta(P_{standard} - P_{base}) = 0.0410, \quad \Delta(P_{TMI} - P_{standard}) = 0.0725.$$

Channels passed 4/4, but circuit-channel cells passed 12/16, pointing to a local circuit issue.

3.4. Test 27: circuit-inverse correctness audit

Test 27 found that the original circuits did not all pass ideal inverse correctness, but the corrected circuits did. The correction was:

$$S \rightarrow Sdg, \quad S^{-1} = S^\dagger.$$

After correction:

$$P_{base} = 0.5791, \quad P_{standard} = 0.6432, \quad P_{TMI} = 0.7576.$$

$$\Delta(P_{standard} - P_{base}) = 0.0641, \quad \Delta(P_{TMI} - P_{standard}) = 0.1144.$$

$$Pr(P_{standard} > P_{base}) = 1.0, \quad Pr(P_{TMI} > P_{standard}) = 1.0, \quad cells = 16/16.$$

4. External Qiskit-Aer visual summary

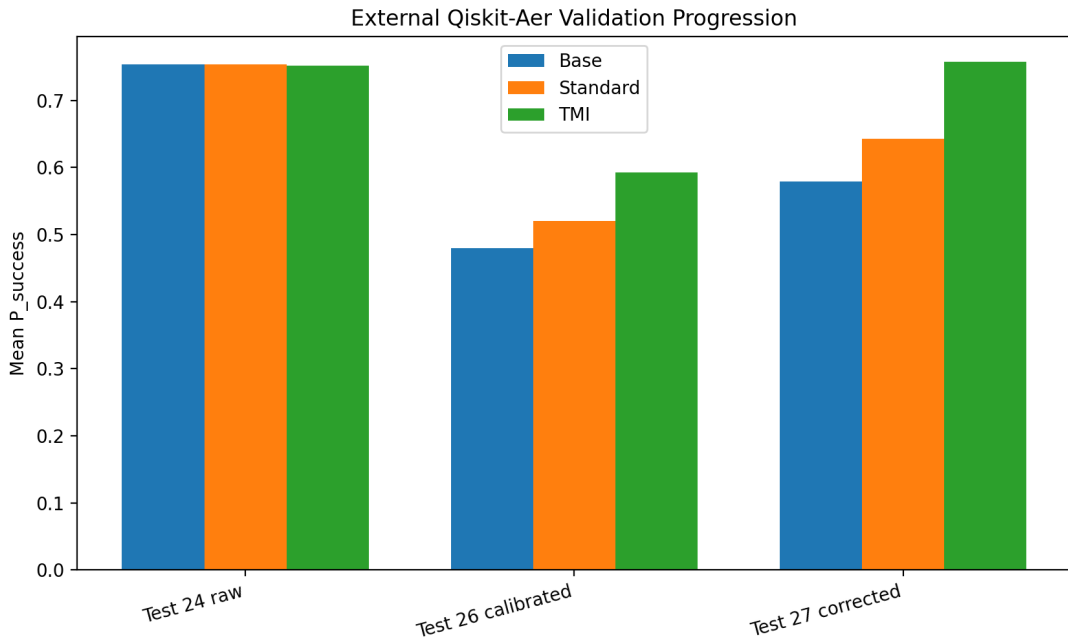


Рис. 1: Mean success probabilities in Tests 24, 26 and 27.

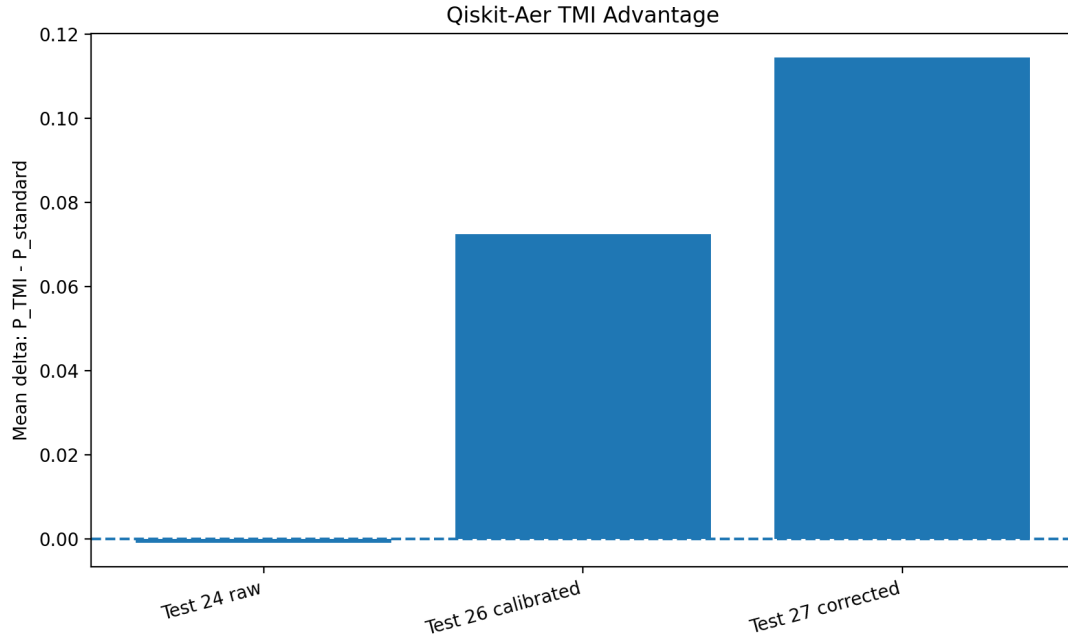


Рис. 2: $P_{TMI} - P_{standard}$: negative in Test 24, positive after calibration/correction.

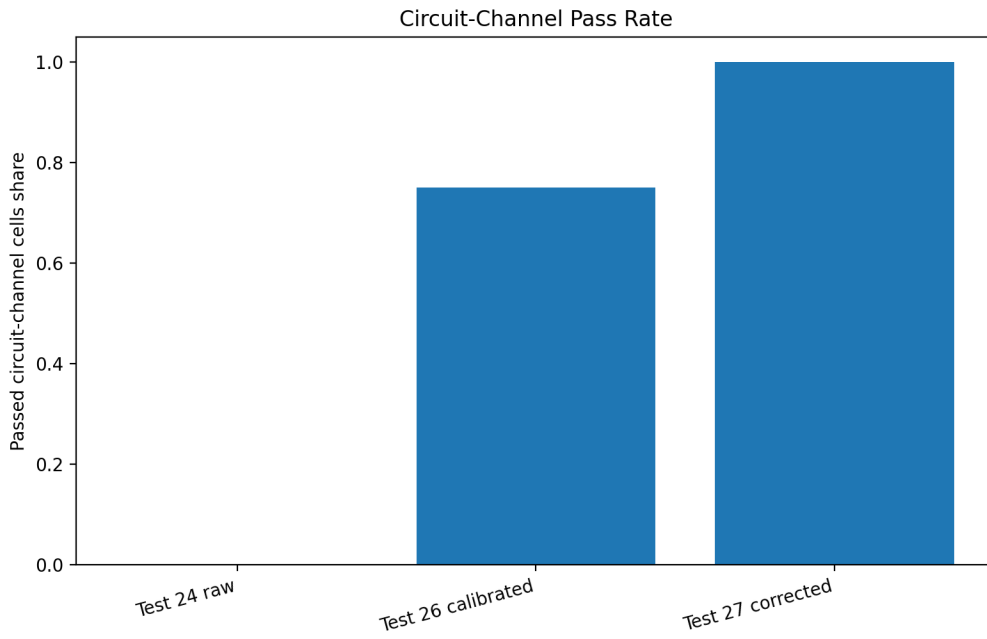


Рис. 3: Passed circuit-channel cells: 0/16, 12/16, 16/16.

5. v4 formulas

$$\text{Calibrate}(I_{sim}) \wedge \text{Correct}(I_{circuit}) \Rightarrow \text{QiskitAerValidation}$$

$$\text{Calibrate}(I_{sim}) \wedge \text{Correct}(I_{circuit}) \Rightarrow P_{TMI} > P_{standard} > P_{base}$$

6. Conclusion

Tests 1-22 established internal computational validation and boundary-manifold control. Tests 23-27 add the external simulator branch. The crucial scientific point is that the report includes the raw external failure, diagnosis, calibration, circuit correction and final corrected validation.

A. Состав пакета

Пакет содержит PDF и LaTeX исходники на русском и английском, JSON/CSV summary, графики внешней Qiskit-Aer ветки, пакеты Tests 12-27 и статусные файлы Tests 24-27.